

Sikao Guo, Ph.D.

RESEARCH SCIENTIST | COMPUTATIONAL BIOPHYSICS | MOLECULAR MODELING

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U.S. Permanent Resident | No sponsorship required

Computational biophysicist with 6+ years developing methods for protein conformational sampling and biomolecular self-assembly. Combines statistical mechanics, molecular simulation, and C++/Python software to test biological mechanisms and deliver validated, reusable research tools.

RESEARCH EXPERIENCE

Inria

Research Engineer

Jul 2025 - Present

Sophia Antipolis, France

- Developed a C++/Python sampler for protein loops, fragments, and intrinsically disordered regions by extending fixed-end loop closure with SE(3) endpoint moves and residue-specific Ramachandran priors.
- Built all-atom validation with side-chain reconstruction, OpenMM minimization, and conformational-quality checks; observed broader exploration than tested generative samplers on the evaluated systems.
- Created [AutoCLIP](#) to generate VMD, PyMOL, and web applications from reusable specifications; deployed **nine structural-biology tools**.

Johns Hopkins University

Assistant Research Scientist (2025); Postdoctoral Fellow (2020-2024)

Jan 2020 - Jun 2025

Baltimore, MD

- Modeled clathrin self-assembly, reproducing experimental kinetics and explaining why lattices fail to stabilize without sufficient adaptor proteins.
- Identified a binding-affinity range that permits HIV Gag/Gag-Pol lattice stability and remodeling on experimentally observed viral-maturation timescales.
- Developed and validated [NERDSS-MPI](#), achieving approximately **90x speedup on 96 CPUs** for 20,000-particle systems while matching serial simulations and analytical benchmarks.
- Automated PDB/mmCIF-to-simulation modeling in [ioNERDSS](#), including ML-assisted affinity initialization; benchmarked **44,000+ structures** and assemblies of 3-720 subunits.

Institute of Physics, Chinese Academy of Sciences

Doctoral Researcher

Sep 2014 - Dec 2019

Beijing, China

- Developed mechanochemical models of kinesin and combined stochastic simulation, molecular dynamics, and umbrella sampling to explain motor coordination and reproduce experimental motility measurements.

TECHNICAL SKILLS

Methods: protein conformational sampling, molecular dynamics, Monte Carlo, reaction-diffusion modeling

Tools: C++, Python, MPI, OpenMM, GROMACS, NumPy, SciPy, Biopython, PyMOL, VMD

EDUCATION

Ph.D. in Physics | Institute of Physics, Chinese Academy of Sciences

2019

B.S. in Physics | Nankai University

2014

SELECTED FIRST-AUTHOR PUBLICATIONS

[Google Scholar](#)

Guo S, Korolija N, et al. [Parallelization of Particle-Based Reaction-Diffusion Simulations Using MPI](#). *Journal of Computational Chemistry* (2025).

Guo S, Saha I, Saffarian S, Johnson ME. [Structure of the HIV Immature Lattice Allows for Essential Lattice Remodeling within Budded Virions](#). *eLife* (2023).

Guo S, Sodt AJ, Johnson ME. [Large Self-Assembled Clathrin Lattices Spontaneously Disassemble without Sufficient Adaptor Proteins](#). *PLOS Computational Biology* (2022).